

Blockchain Technology

A distributed, tamper-resistant digital ledger

068

Blockchain Technology is a distributed digital ledger that records transactions across many computers in a way that makes altering past records extremely difficult, underpinning cryptocurrencies and other applications.

DEFINITION

A distributed, decentralized ledger technology that records transactions across a network of computers in a secure, tamper-resistant manner.

CORE CONCEPTS

- Distributed ledger
- Consensus mechanisms
- Cryptographic hashing
- Smart contracts

WHY IT MATTERS

Blockchain enables trust between parties without needing a central authority, opening new possibilities for finance, supply chains, and digital ownership verification.

APPLICATIONS

- Cryptocurrency transactions
- Supply chain traceability
- Smart contract automation
- Digital identity verification

REAL-LIFE EXAMPLES

- Bitcoin and other cryptocurrencies
- Supply chain tracking for food safety
- Non-fungible tokens (NFTs) for digital ownership

WORKING PRINCIPLE

INPUT / PROBLEM

CORE PROCESS

OUTCOME / APPLICATION

Transactions are grouped into blocks, cryptographically linked to previous blocks, and verified by a network of computers using a consensus mechanism, making the resulting chain extremely difficult to alter retroactively.

ADVANTAGES

- Removes need for a trusted central intermediary
- Provides transparent, tamper-resistant record-keeping
- Enables new forms of digital ownership and automation

LIMITATIONS

- Can be energy-intensive depending on consensus method
- Transaction speeds can be slower than traditional systems
- Regulatory frameworks are still evolving

QUICK FACTS

- Bitcoin, the first blockchain application, was introduced in a 2008 paper by the pseudonymous Satoshi Nakamoto
- Some blockchain networks can process thousands of transactions per second using newer consensus methods

CAREER OPPORTUNITIES

Blockchain developer, Smart contract engineer, Cryptocurrency analyst, Blockchain security auditor

INDUSTRY APPLICATIONS

Finance, Supply chain and logistics, Healthcare records, Digital art and media

RESEARCH AREAS

Energy-efficient consensus mechanisms, Blockchain scalability, Cross-chain interoperability

FUTURE SCOPE

Beyond cryptocurrency, blockchain is expanding into supply chain, healthcare records, and digital identity applications.

KEY TERMS

Distributed ledger

Smart contract

Consensus mechanism

Cryptographic hash

CURRENT TRENDS

Growing focus on more energy-efficient consensus mechanisms as environmental concerns around blockchain energy use persist.

SUMMARY

Blockchain technology creates tamper-resistant digital records without a central authority, powering cryptocurrency and expanding into many new applications.